

OpenClaw AI Weather Station Tutorial

Before starting, please complete [01-Peripheral hardware configuration], [02-openclaw API-KEY configuration], [03-openclaw switching model], [04-OpenClaw prompt words], [05-AI voice interaction configuration], [06-Three dialogue scheme configuration tests] in the `Preparation before use` directory. If you are currently only using TUI/WhatsApp, you can skip `05` first. This article assumes that the basic environment has been prepared and only the AI weather station project itself will be launched.

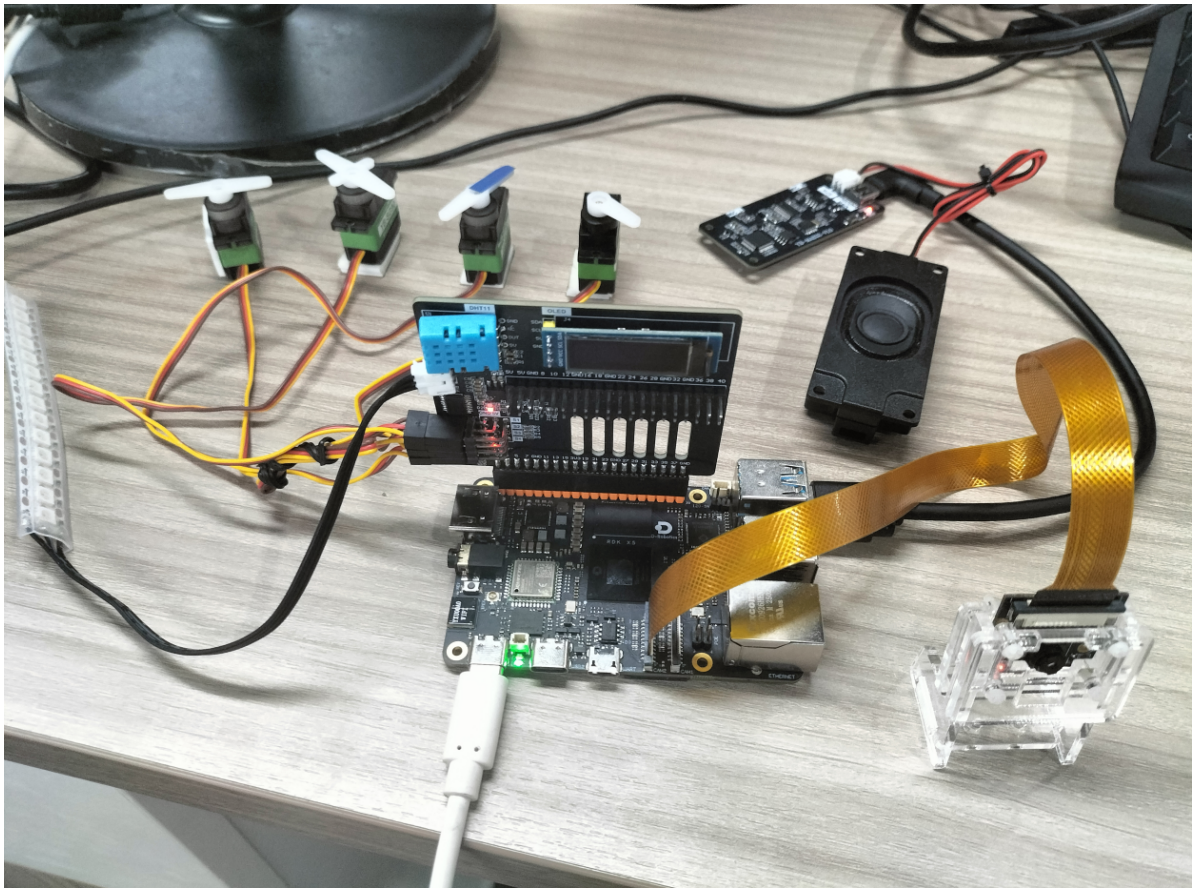
1. Tutorial Objectives

This article demonstrates the capabilities of OpenClaw in a desktop weather station scenario through temperature and humidity collection, information display, and AI interpretation. After completing this tutorial, readers will be able to:

- Learn how to combine DHT, OLED and RGB to create a local AI weather station
- Converse with OpenClaw through WhatsApp, TUI, and voice code
- Complete four project scenarios: environment collection, AI broadcast, weather card, status warning

2. Hardware preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip/RGB lamp bead module
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software preparation

Code entry that will be used in the tutorial

- Temperature and humidity reading command: `/home/sunrise/hardware_ha1_demo/dht`
- OLED control command: `/home/sunrise/hardware_ha1_demo/oled`
- RGB control command: `/home/sunrise/hardware_ha1_demo/rgb`
- Voice entrance: `/home/sunrise/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/sunrise/voice_to_openclaw/src/text_main.py`

The current focus of this set of tutorials is on the "local environmental weather station", which is based on DHT's indoor temperature and humidity data for display and explanation;

4. Test the basic capabilities of the project

Before starting the current project, please complete a basic dialogue self-test of `openclaw tui`; if you have not done so yet, please read [06-Three dialogue scheme configuration tests] first. After passing, enter the AI weather station special test.

It is recommended to enter OpenClaw TUI first:

```
openclaw tui
```

```
sunrise@ubuntu:~/./openclaw$ openclaw tui
♥ OpenClaw 2026.3.24 (cff6dc9) - You had me at 'openclaw gateway start.'
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main

hello

你好呀，主人！☺ 我是小亚，随时待命。

有什么需要我帮忙的吗？
gateway connected | idle
agent main | session main (openclaw-tui) | deepseek/deepseek-v4-pro | tokens 12k/1.0m (1%)
```

Then test the following types of instructions in sequence:

- Read current temperature and humidity
- Display weather cards on OLED
- If the environment is normal, the green light will turn on
- Give me a brief environmental announcement

5. Three options for talking to OpenClaw

Solution	Suitable scenario	Advantages	Suggested positioning
WhatsApp	Remote viewing of the environment, classroom demonstrations	Suitable for remotely asking "How is the current environment?"	Showing the "Remote Weather Station Assistant"
TUI	Local debugging, threshold verification	The easiest way to troubleshoot sensors and display logic	Run through the collection link first
Voice code	Voice broadcast, desktop assistant	Closer to daily use experience	As the final demonstration solution

If the AI voice module + speaker is optional, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to open broadcast parameters

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter is to control whether to broadcast. After modification, save it and run the following program on the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp solution

For WhatsApp access and robot creation steps, please refer directly to [06-Configuration Test of Three Dialogue Schemes]. After it is created, you can remotely view the environment status on WhatsApp, for example:

- How is the indoor environment now? If it's too hot, remind me.

20:34



< 2



+86 153 3885 7526 (你)

给自己发消息

...waiting for me...

20:33 ✓✓

How is the indoor environment now? If it is too hot, remind me.

20:34 ✓✓

Current indoor environment readings:



Temperature: 29.7°C (85.5°F)



Humidity: 54%

Assessment: The temperature is **just below the hot threshold** (30°C). It's warm but not excessively hot yet.

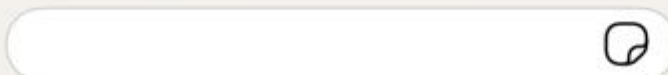
Recommendation:

- The room is comfortably warm, but if the temperature rises above 30°C, I'll remind you with a red light warning
- Current humidity at 54% is in a comfortable range (40-60% is ideal)
- Consider ventilation if you feel warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.

20:34 ✓✓



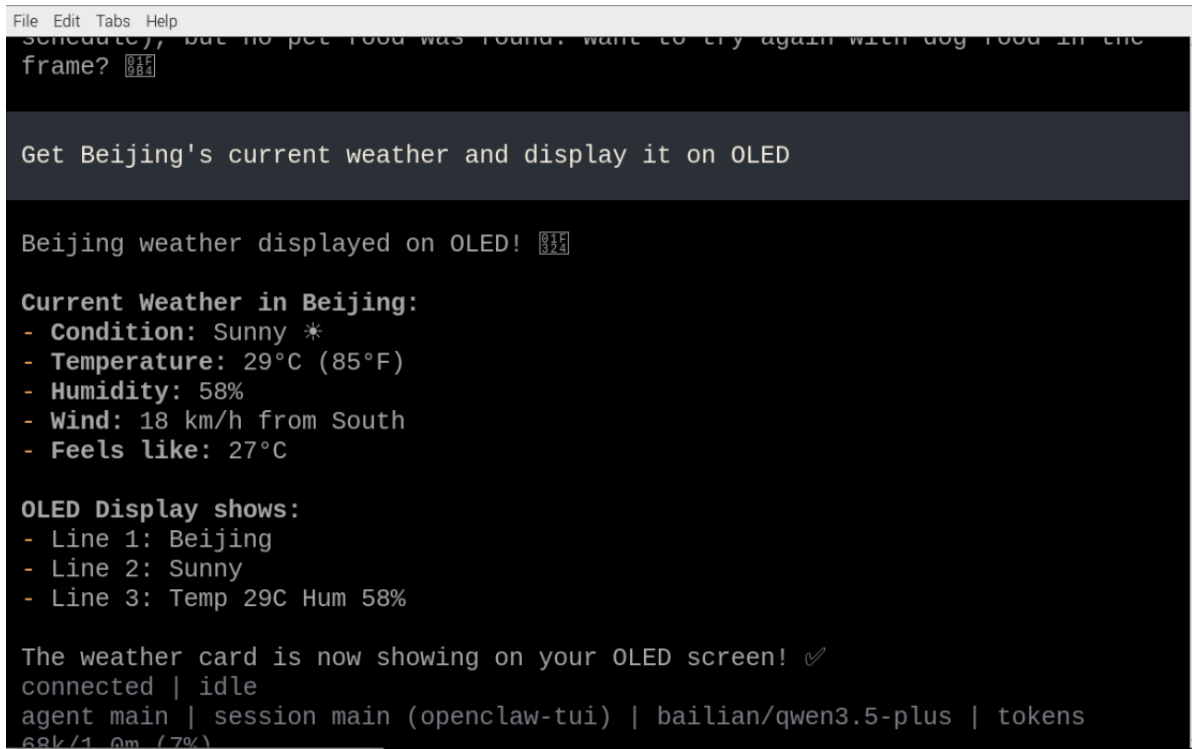
5.2 TUI scheme

If you have completed the TUI self-test in Part 0, enter the terminal text dialogue directly:

```
openclaw tui
```

Then enter project-related instructions, such as:

- Get the current weather in Beijing and display it on OLED



```
File Edit Tabs Help
Schedule, but no pet food was found. Want to try again with dog food in the
frame? [0.1]

Get Beijing's current weather and display it on OLED

Beijing weather displayed on OLED! [0.1]

Current Weather in Beijing:
- Condition: Sunny *
- Temperature: 29°C (85°F)
- Humidity: 58%
- Wind: 18 km/h from South
- Feels like: 27°C

OLED Display shows:
- Line 1: Beijing
- Line 2: Sunny
- Line 3: Temp 29C Hum 58%

The weather card is now showing on your OLED screen! ✓
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
68k/1.0m (7%)
```

5.3 Voice code scheme

```
Requires optional ai voice interaction module + speaker
```

For voice module installation, environment configuration and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After waking up, you can directly say the control content, such as:

- ai weather station

```

[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号，开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音，正在转写

[你] Ai weather station.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## 🌤️AI Weather Station Complete!

**Local Sensor Readings:**
-
-           **Indoor Temperature:** 30°C (86°F)
- **Indoor Humidity:** 51%
-           **Outdoor Weather (Shenzhen):**
- **Condition:** Partly cloudy 🌤️
-
-           **Outdoor Temperature:** 28°C (82°F)
- **Feels Like:** 35°C (95°F)
-
-           **Outdoor Humidity:** 84%
- **Wind:** 9 km/h from South
- **Rain
    Chance:** 72% today

**3-Day Forecast:**
- **Today (May 14):** High
    28°C, Low 23°C - Patchy rain nearby
- **Tomorrow (May 15):** High
    26°C, Low 23°C - Moderate rain expected

```

6. Project implementation

Note: The following are all dialogue demonstrations in `openc1aw_tui` mode. You can also choose WhatsApp or voice mode to complete it.

6.1 Scenario 1: Environment collection

Experimental goals

Read the temperature and humidity of the local environment to complete the most basic data sampling of the weather station.

Effect description

OpenClaw will call DHT to obtain the current environmental data and return the results to the user to form a real-time "weather reading" of the indoor environment.

Example instructions

- Read the current ambient temperature and humidity
- What is the indoor temperature and humidity now?
- Help me check the current weather data

6.2 Scenario 2: AI broadcast

Experimental goals

Let OpenClaw not only report numbers, but also summarize the state of the environment like a weather announcer.

Effect description

OpenClaw can summarize "26 degrees, 57% humidity" into more natural words, such as "the environment is more comfortable", "a bit stuffy", "the air is slightly dry", etc.

Example instructions

- Give me a brief environmental announcement
- Is the environment comfortable now?
- Make a weather summary based on current readings

6.3 Scenario 3: OLED weather card

Experimental goals

Display the local temperature and humidity results as a page more like a "weather card".

Effect description

In addition to text-only results, OLED can also arrange location, status, temperature and humidity into a concise card, making the entire project more like a displayable desktop device.

Example instructions

- Display an indoor weather card on OLED
- Get the current weather in xx place and display it on OLED

6.4 Scenario 4: RGB status warning

Experimental goals

Use RGB colors to quickly express the current environment status.

Effect description

For example, it lights up green when the environment is comfortable, yellow when it is hot, and red when it is abnormal. In this way, even without reading the text, you can quickly know the current state of the environment.

Example instructions

- If the environment is normal, the green light will turn on
- If the temperature is high, the yellow light will turn on
- If there is an abnormality, the red light will flash.